Sham Satish Thakare

Computer Science Researcher · *Reliable and Adaptive Intelligent Systems*

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RESEARCH STATEMENT

I conduct research on **Reliable and Adaptive Intelligent Systems**, focusing on uncertainty calibration in language-model post-training (RLVR credit assignment, reasoning complexity confounding), pre-hoc plasticity diagnostics for foundation model scheduling, multi-stage agent tool safety, failure-domain aware consensus quorum adaptation in C++20, and microservice causal observability. I prioritize rigorous statistical evaluation protocols and fully reproducible open-source artifacts.

EDUCATION

Anantrao Pawar College of Engineering and Research (APCOER), Pune, India **01/2020 – 06/2024**

Bachelor of Technology (B.Tech.) in Artificial Intelligence and Data Science | **Cumulative GPA: 8.70 / 10.00**

- **B.Tech. Research Thesis:** *Machine Learning Diagnostic Evaluation Protocols & Statistical Algorithms*
- **Thesis Advisers:** Prof. Sagar Gawande and Prof. Sunil Thakare
- **Mathematics Coursework:** Advanced Linear Algebra, Probability & Random Processes, Multivariate Calculus, Discrete Mathematics, Numerical Methods.
- **Computer Science Coursework:** Machine Learning, Deep Learning, Reinforcement Learning, Distributed Systems, Operating Systems, Database Management Systems, Computer Networks, Data Structures & Algorithms.

RESEARCH EXPERIENCE

Independent Computer Science Researcher — *Reliable Adaptive Intelligent Systems* **06/2024 – Present**

Uncertainty-Weighted Credit Assignment in LM RL (Flagship Study) | 01/2026 – Present

- **PROBLEM:** Investigated whether token predictive entropy validly measures reasoning uncertainty in Group Relative Policy Optimization (GRPO) post-training, or operates as a proxy for sequence length.
- **METHOD:** Formulated PyTorch diagnostic probes and a 5-way RL benchmark evaluated across untouched GSM8K mathematical prompts (N=100) with Qwen2.5-0.5B-Instruct.
- **EVIDENCE:** Uncovered strong positive correlation between token entropy and completion length (r = 0.486, 95% CI [+0.318, +0.627]), which collapsed to non-significance ($r_{partial}$ = -0.092, p = 0.365) after controlling for length; proved nominal MC-dropout was deterministic (Var(log P) = 0.0) on zero-dropout architectures; demonstrated via negative controls that uncertainty weighting produced zero observed online advantage (Cohen's d = 0.00, 80.00% ± 0.00% Pass@1) over outcome-supervised GRPO across N=3 seeds.
- **CONTRIBUTION:** Released open-source PyTorch probing suite ear_grpo_reasoning and canonical research essay establishing diagnostic protocols for RLVR credit assignment.

CARLS: Foundation Model RL Plasticity Prediction | 02/2026 – Present

- **PROBLEM:** Addressed computational overhead in executing full policy-gradient RL loops to select optimal pre-trained language model checkpoints for post-training alignment.
- **METHOD:** Formulated representation geometry diagnostic probes analyzing gradient norms and feature drift across Pythia and Qwen checkpoint series.
- **EVIDENCE:** Predicted reward plasticity pre-hoc, reducing RL training compute FLOPs by 18.7% while preserving post-training Pass@4 accuracy.
- **CONTRIBUTION:** Developed open-source RL plasticity benchmarking library adaptive-rl-forge.

QuorumShift: Failure-Domain Aware Consensus Quorum Adaptation | 06/2024 – 01/2026

- **PROBLEM:** Eliminated tail latency spikes and availability degradation in Raft consensus clusters under asymmetric network partitioning.
- **METHOD:** Designed a C++20 dynamic quorum adaptation algorithm that dynamically adjusts voting thresholds and replica selection using real-time latency profiles.
- **EVIDENCE:** Reduced 99th percentile write latency under asymmetric network fault injection while maintaining strict linearizability.
- **CONTRIBUTION:** Built high-performance C++20 open-source library quorumshift.

Undergraduate Researcher — APCOER Pune **02/2023 – 05/2024**

Advisers: Prof. Sagar Gawande and Prof. Sunil Thakare

- Formulated machine learning diagnostic evaluation frameworks and statistical algorithms for high-dimensional data evaluation.
- Conducted empirical statistical evaluation protocols and benchmarked model convergence across synthetic and real-world datasets.

SELECTED PUBLICATIONS & MANUSCRIPTS

- **[Flagship Manuscript] Thakare, S. S.** "When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LLMs." *Working Paper / Research Note*, 2026. [PDF Paper] [Code] [Research Essay]
  Contribution: Exposes completion-length confounding (r = 0.486, N=100) in token predictive entropy, proves MC-dropout determinism on zero-dropout LLMs, and proves via negative controls that consistency-weighted GRPO provided zero observed online advantage (d = 0.00, N=3 seeds) over outcome-supervised GRPO.

- **[RL Plasticity] Thakare, S. S.** "Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints: A Cross-Architecture Diagnostic Study." *Working Paper*, 2026. [Code]
  Contribution: Formulates gradient norm and feature drift diagnostic probes across Pythia/Qwen checkpoint series, predicting reward plasticity prior to RL training and reducing FLOPs by 18.7%.

- **[Distributed Consensus] Thakare, S. S.** "AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus." *Working Paper*, 2026. [Code]
  Contribution: C++20 dynamic quorum adaptation engine for Raft consensus clusters eliminating tail latency under asymmetric network partitioning.

- **[Error Recovery Protocols] Thakare, S. S.** "recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning." *Working Paper*, 2026. [PDF Paper] [Code]

- **[Agent Tool Safety] Thakare, S. S.** "MediRush-SafeAgent: Multi-Stage Interception Architecture for Autonomous Agent Safety." *Research Project*, 2026. [Code]

- **[Cloud Systems Causal Observability] Thakare, S. S.** "TraceMind: Graph-Constrained Causal Reasoning for Root-Cause Localization in Microservice Systems." *Working Paper*, 2026. [Code]

- **[Hardware Enclave Confidentiality] Thakare, S. S.** "EnclaveShield: Zero-Knowledge Memory Attestation and Side-Channel Mitigation for Hardware Enclaves." *Working Paper*, 2026. [Code]

PROFESSIONAL EXPERIENCE

Spectra Corporate Service, Pune, India **06/2024 – Present**

Data Engineer / Machine Learning Specialist

- Engineered high-throughput parallel data processing pipelines and cloud telemetry infrastructure using C++, Python, and SQL.
- Optimized multithreaded cloud workflow automation scripts and telemetry data pipelines, accelerating execution latency across compute clusters.
- Implemented automated schema verification, metric logging, and performance benchmarking across enterprise data services.

TEACHING EXPERIENCE

APCOER Pune (Department of AI & Data Science), Pune, India **06/2023 – 05/2024**

Teaching Assistant

- Provided instructional support and doubt resolution for undergraduate courses in *Data Structures*, *Algorithms*, and *Foundations of Machine Learning*.
- Conducted lab sessions, graded programming assignments, and developed automated grading scripts in Python and C++.

HONORS & AWARDS

- **Gold Medal, Sprint / Distance Running** — Inter-College Athletic Meet, 2023.
- **APCOER Academic Excellence Distinction** — Department of AI & Data Science (CGPA: 8.70 / 10.00), 2024.

OPEN SOURCE RESEARCH SOFTWARE

- **ear_grpo_reasoning**: PyTorch probing suite for token predictive entropy, MC-dropout, and GRPO advantage scaling.
- **adaptive-rl-forge**: Diagnostic probe library for foundation model checkpoint reward plasticity.
- **quorumshift**: High-performance C++20 dynamic quorum adaptation for Raft consensus clusters.
- **medirush-safeagent**: Multi-stage runtime policy interception architecture for agent tool safety.
- **recovery_eval**: State-contrast continuation evaluation framework for neural reasoning error recovery.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++ (C++20), SQL, Java, C#/.NET, JavaScript/TypeScript.
- **Machine Learning & AI:** PyTorch, Hugging Face Transformers, GRPO, RLVR, scikit-learn, NumPy, Pandas, Matplotlib, SciPy.
- **Systems & Backend:** C++20 Multithreading, Distributed Consensus (Raft), OpenTelemetry, Microservices, REST APIs, POSIX.
- **Tools & Infrastructure:** Linux, Git, Docker, GitHub Actions, LaTeX, PyTest, Google Cloud Platform.
- **Research Methods:** Experimental Design, Partial Correlation Analysis, AUROC Evaluation, Multi-Seed Controls, Reproducibility Protocols.